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Lab - Using Windows PowerShell

Objectives

The objective of the lab is to explore some of the functions of PowerShell.

Part 1: Access PowerShell console.

Part 2: Explore Command Prompt and PowerShell commands.

Part 3: Explore cmdlets.

Part 4: Explore the netstat command using PowerShell.

Part 5: Empty recycle bin using PowerShell.

Background / Scenario

PowerShell is a powerful automation tool. It is both a command console and a scripting language. In this lab, you will use the console to execute some of the commands that are available in both the command prompt and PowerShell. PowerShell also has functions that can create scripts to automate tasks and work together with the Windows Operating System.

Required Resources

- 1 Windows PC with PowerShell installed and internet access

Instructions

Part 1: Access PowerShell console.

- Click **Start**. Search and select **powershell**.
- Click **Start**. Search and select **command prompt**.

Part 2: Explore Command Prompt and PowerShell commands.

- Enter **dir** at the prompt in both windows.

What are the outputs to the **dir** command?

Kết quả **dir** giống nhau ở cả hai cửa sổ — liệt kê danh sách file và thư mục con trong thư mục hiện tại (tên, kích thước, ngày sửa đổi) nhưng bên PowerShell có thêm mục mode (hiển thị quyền)

- Try another command that you have used in the command prompt, such as **ping**, **cd**, and **ipconfig**.

What are the results?

Các lệnh **ping**, **cd**, **ipconfig** chạy được và cho kết quả giống nhau ở cả Command Prompt lẫn PowerShell, vì PowerShell tương thích ngược với các lệnh cmd truyền thống.

Part 3: Explore cmdlets.

- PowerShell commands, cmdlets, are constructed in the form of *verb-noun* string. To identify the PowerShell command to list the subdirectories and files in a directory, enter **Get-Alias dir** at the PowerShell prompt.

```
PS C:\Users\CyberOpsUser> Get-Alias dir
```

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Aliasdir -> Get-ChildItem

What is the PowerShell command for **dir**?Lệnh PowerShell tương ứng với **dir** là **Get-ChildItem** (dir chỉ là alias của nó).

- b. For more detailed information about cmdlets, perform an internet search for **Microsoft powershell cmdlets**.
- c. Close the Command Prompt window when done.

Part 4: Explore the netstat command using PowerShell.

- a. At the PowerShell prompt, enter **netstat -h** to see the options available for the **netstat** command.

PS C:\Users\CyberOpsUser> **netstat -h**

Displays protocol statistics and current TCP/IP network connections.

NETSTAT [-a] [-b] [-e] [-f] [-n] [-o] [-p proto] [-r] [-s] [-x] [-t] [interval]

-a Displays all connections and listening ports.

-b Displays the executable involved in creating each connection or listening port. In some cases well-known executables host multiple independent components, and in these cases the sequence of components involved in creating the connection or listening port is displayed. In this case the executable name is in [] at the bottom, on top is the component it called, and so forth until TCP/IP was reached. Note that this option can be time-consuming and will fail unless you have sufficient permissions.

<some output omitted>

- b. To display the routing table with the active routes, enter **netstat -r** at the prompt.

PS C:\Users\CyberOpsUser> **netstat -r**

=====

= Interface List

3...08 00 27 a0 c3 53Intel(R) PRO/1000 MT Desktop Adapter

10...08 00 27 26 c1 78Intel(R) PRO/1000 MT Desktop Adapter #2

1.....Software Loopback Interface 1

=====

IPv4 Route Table

=====

= Active Routes:

Network Destination Netmask Gateway Interface Metric

0.0.0.0 0.0.0.0 192.168.1.1 192.168.1.5 25

127.0.0.0 255.0.0.0 On-link 127.0.0.1 331 127.0.0.1 255.255.255.255

On-link 127.0.0.1 331 127.255.255.255 255.255.255.255 On-link 127.0.0.1

331 169.254.0.0 255.255.0.0 On-link 169.254.181.151 281 169.254.181.151

255.255.255.255 On-link 169.254.181.151 281 169.254.255.255

255.255.255.255 On-link 169.254.181.151 281 192.168.1.0 255.255.255.0

On-link 192.168.1.5 281 192.168.1.5 255.255.255.255 On-link 192.168.1.5

```
281 192.168.1.255 255.255.255.255 On-link 192.168.1.5 281
```

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```
224.0.0.0 240.0.0.0 On-link 127.0.0.1 331 224.0.0.0 240.0.0.0 On-link
192.168.1.5 281 224.0.0.0 240.0.0.0 On-link 169.254.181.151 281
255.255.255.255 255.255.255.255 On-link 127.0.0.1 331 255.255.255.255
255.255.255.255 On-link 192.168.1.5 281 255.255.255.255 255.255.255.255
On-link 169.254.181.151 281
```

Persistent Routes:

None

IPv6 Route Table

= Active Routes:

If Metric Network Destination Gateway

1 331 ::1/128 On-link

3 281 fe80::/64 On-link

10 281 fe80::/64 On-link

10 281 fe80::408b:14a4:7b64:b597/128

On-link

3 281 fe80::dd67:9e98:9ce0:51e/128

On-link

1 331 ff00::/8 On-link

3 281 ff00::/8 On-link

10 281 ff00::/8 On-link

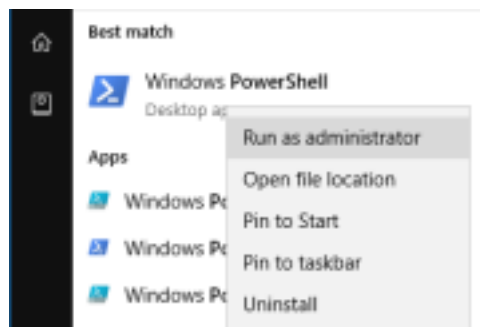
= Persistent Routes:

None

What is the IPv4 gateway?

IPv4 gateway là **192.168.1.1**

- c. Open and run a second PowerShell with elevated privileges. Click **Start**. Search for PowerShell and right click **Windows PowerShell** and select **Run as administrator**. Click **Yes** to allow this app to make changes to your device.



- d. The netstat command can also display the processes associated with the active TCP connections. Enter the **netstat -abno** at the prompt.

```
PS C:\Windows\system32> netstat -abno
```

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```

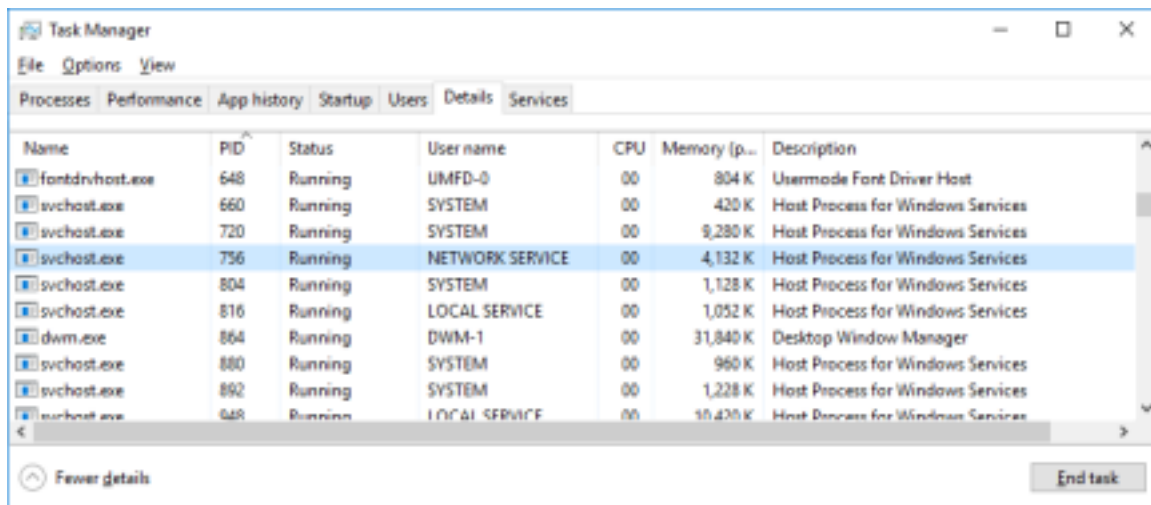
Proto Local Address Foreign Address State PID
LISTENING 756 RpcSs
[svchost.exe]
TCP 0.0.0.0:445 0.0.0.0:0 LISTENING 4 Can not obtain ownership
information
TCP 0.0.0.0:49664 0.0.0.0:0 LISTENING 444 Can not obtain ownership
information
TCP 0.0.0.0:49665 0.0.0.0:0 LISTENING 440 Schedule
[svchost.exe]
TCP 0.0.0.0:49666 0.0.0.0:0 LISTENING 304 EventLog
[svchost.exe]
TCP 0.0.0.0:49667 0.0.0.0:0 LISTENING 1856 [spoolsv.exe]
TCP 0.0.0.0:49668 0.0.0.0:0 LISTENING 544 <some output omitted>

```

e. Open the Task Manager. Navigate to the **Details** tab. Click the **PID** heading so the PID are in order. f.

Select one of the PIDs from the results of netstat -abno. PID 756 is used in this example.

g. Locate the selected PID in the Task Manager. Right-click the selected PID in the Task Manager to open the **Properties** dialog box for more information.



What information can you get from the Details tab and the Properties dialog box for your selected PID? Tab Details và hộp thoại Properties cho biết: tên tiến trình, PID, tên người dùng chạy tiến trình, mức sử dụng CPU/bộ nhớ, mô tả tiến trình, đường dẫn file thực thi, ngày tạo, chữ ký số (digital signature) của file

Part 5: Empty recycle bin using PowerShell.

PowerShell commands can simplify management of a large computer network. For example, if you wanted to implement a new security solution on all servers in the network you could use a PowerShell command or script to implement and verify that the services are running. You can also run PowerShell commands to simplify actions that would take multiple steps to execute using Windows graphical desktop tools.

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- Open the Recycle Bin. Verify that there are items that can be deleted permanently from your PC. If not, restore those files.
- If there are no files in the Recycle Bin, create a few files, such as text file using Notepad, and place them into the Recycle Bin.
- In a PowerShell console, enter **clear-recyclebin** at the prompt.

```
PS C:\Users\CyberOpsUser> clear-recyclebin
```

```
Confirm
```

```
Are you sure you want to perform this action?
```

```
Performing the operation "Clear-RecycleBin" on target "All of the contents of the Recycle Bin".
```

```
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "Y") : y
```

What happened to the files in the Recycle Bin?

Các file trong Recycle Bin bị xóa vĩnh viễn khỏi hệ thống, không thể khôi phục lại.

Reflection Question

PowerShell was developed for task automation and configuration management. Using the internet, research commands that you could use to simplify your tasks as a security analyst. Record your findings.

